

VALAY BUNDELE



RESEARCH INTERESTS

Object Tracking, Self-supervised Learning, Efficient Transformers, Digital Pathology, Surgical Video Analysis

EDUCATION

International Max Planck Research School for Intelligent Systems

Tuebingen, Germany

PhD student in the Computer Graphics and Vision lab at University of Tuebingen

Sept 2023 – Present

- Thesis (tentative): Robust Tracking Through Occlusions
- Advisor: [Prof. Hendrik Lensch](#)

Indian Institute of Technology Bombay

Mumbai, India

B.Tech in Energy Engineering with M.Tech in AI and Data Science

July 2018 – July 2023

- Cumulative GPA: **8.77/10.0**
- Minor Degree in Computer Science and Engineering

Aklank Public School(CBSE)

Kota, India

Intermediate/+2

Percentage: 91.40%

PUBLICATIONS

** denotes equal contribution*

1. **V. Bunde**le, S. Agrawal, M. Hosseinzadeh, N. Nguyen, H. Lensch; ‘When Trackers Fail: VLM-Guided Verification and Recovery for Robust Video Object Segmentation’, Advances in Neural Information Processing Systems (NeurIPS) 2026
2. A. Stammer, **V. Bunde**le, M. Hosseinzadeh, H. Lensch; ‘[Learning Where to Focus: Self-Supervised Multi-Scale ViTs for Histopathology](#)’, DAGM German Conference on Pattern Recognition (GCPR) 2026
3. **V. Bunde**le, M. Hosseinzadeh, H. Lensch; ‘[ReMeDI: Refined Memory for Disambiguation of Identities with SAM3 in Surgical Segmentation](#)’, International Conference on Medical Image Computing And Computer Assisted Intervention (MICCAI) 2026
4. **V. Bunde**le*, K. Saritaş*, B. Kargi*, O. Çal*, K. Tezören*, Z. Ghaderi, H. Lensch; ‘[Evaluating Self-Supervised Learning in Medical Imaging: A Benchmark for Robustness, Generalizability, and Multi-Domain Impact](#)’, British Machine Vision Conference (BMVC) 2025
5. E. Grosskopf*, **V. Bunde**le*, M. Hosseinzadeh*, H. Lensch; ‘[HistDiST: Histopathological Diffusion-based Stain Transfer](#)’, DAGM German Conference on Pattern Recognition (GCPR) 2025
6. **V. Bunde**le, M. Hosseinzadeh, H. Lensch; ‘[You Are Your Best Teacher: Semi-Supervised Surgical Point Tracking with Cycle-Consistent Self-Distillation](#)’, SynData4CV Workshop at Computer Vision and Pattern Recognition (CVPR) 2025
7. P. Bele, **V. Bunde**le, A. Bhattacharya, A. Jha, G. Roig, B. Banerjee; ‘[Learning Class and Domain Augmentations for Single-Source Open-Domain Generalization](#)’, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2024
8. **V. Bunde**le, M. Bhupati, B. Banerjee, A. Grover; ‘[Scaling Vision-and-Language Navigation with Sub-Optimal Trajectories](#)’, Transactions of Machine Learning Research (TMLR) 2024
9. S. Nasser, S. Shamsi, **V. Bunde**le, B. Garg, A. Sethi; ‘[Perceptual cGAN for MRI Super-resolution](#)’, IEEE Engineering in Medicine and Biology Conference (EMBC) 2022

10. D. Pal, **V. Bundele**, R. Sharma, B. Banerjee, Y. Jeppu; ‘[Few-Shot Open-Set Recognition of Hyperspectral Images with Outlier Calibration Network](#)’, IEEE Winter Conference on Applications of Computer Vision (WACV) 2022
11. D. Pal*, **V. Bundele***, B. Banerjee, Y. Jeppu; ‘[SPN: Stable Prototypical Network for Few-Shot Learning-Based Hyperspectral Image Classification](#)’, IEEE Geoscience and Remote Sensing Letters, 19, 1-5.

PROFESSIONAL AND TEACHING EXPERIENCE

Software Development Intern | Amazon Web Services

May – July 2021

Interned in the Aurora CP team which works on AWS Relational Database Management System

- Worked on graph-based **integrated testing** framework of Relational Database Service to detect destructive paths
- Utilized **Java-callgraph** to generate a static call graph and thereby get the function calls made in each function
- Implemented a **depth-first search** based approach to recursively check for annotation on each function using **Reflection API** and classified paths having function calls with a specific annotation as destructive

Teaching Assistantships

Facilitating smooth course organization, grading papers, mentoring students, conducting tutorials and help sessions

- **TA:** Computer Graphics, Prof. Hendrik Lensch Oct - Mar 2025
- **TA:** Practical Course for Vision and NLP, Prof. Hendrik Lensch Apr - July 2024
- **TA:** GNR 638 - Machine Learning for Remote Sensing II, Prof. Biplab Banerjee Aug - Nov 2021
- **Head TA:** GNR 638 - Machine Learning for Remote Sensing II, Prof. Biplab Banerjee Aug - Nov 2022

Academic Reviewing

Peer-reviewer for top-tier conferences and journals in computer vision and machine learning

- Reviewer for: NeurIPS, CVPR, ECCV, WACV, BMVC, MICCAI, TMLR, Pattern Recognition

TECHNICAL SKILLS

Programming Languages: C++, Python

Libraries/Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Numpy, Pandas

Software: Microsoft Office, L^AT_EX, Git, VS Code

RELEVANT COURSEWORK

Machine Learning: Introduction to Machine Learning, Advanced ML, Medical Image Computing, Foundations of Intelligent and Learning Agents, Learning with Graphs, Deep Learning : Theory and Practice

Computer Science: Operating Systems, Logic in CS, Data Structures & Algorithms, Design & Analysis of Algorithms

Mathematics: Calculus, Linear Algebra, Differential Equations, Numerical Analysis

SELECT MENTORING AND LEADERSHIP ROLES

Software Subsystem Head | Team Rakshak

Apr 2021 - July 2022

Head of the software subsystem of Team Rakshak which works on design and development of UAVs

- Led and managed a team of **10 members** to develop algorithms for the tasks of Object Detection, Object Recognition, and Mapping of surveyed areas for **AUVSI SUAS** competition held in **Maryland, USA**
- Employed YOLOv5 for detection and trained classifiers having **ResNet** backbone to identify object characteristics
- Implemented **PCA** to find the orientation of objects and performed image stitching using **SIFT** to generate a map

Coordinator at Mood Indigo, IIT Bombay | Competitions and LYP

Aug - Dec 2019

Asia's largest college cultural festival; 146,000+ foot-fall with 240+ events

- Member of the team that ideated and revamped the governing rules of **5+** competitions in various genres
- Assisted in conceptualizing and organizing Multicity Competitions in **5+** cities, thus increased outreach
- Invited renowned artists across various cultural genres as judges for the competitions on a **strict zero budget**

ACADEMIC ACHIEVEMENTS AND EXTRACURRICULAR ACTIVITIES

Achievements	● Secured All India Rank 2162 in Joint Entrance Examination among 1.1 million candidates (2018) ● Among Top 1% students (nationally) in first stage of IJSO among 46,688 candidates (2014-15) ● Secured All India Rank 18 in STaRT examination conducted by Resonance Eduventures (2015) ● Obtained International Rank 193 in National Science Olympiad conducted by SOF (2013)
Mentorship	● Mentored 2 students at Summer Of Science 2020 to understand concepts of Deep Learning ● Guided 8 students on the project titled "Image Colorization" - IITB Seasons of Code 2021 ● Mentored 5 students on the project titled "Image Editing" - IITB Seasons of Code 2022
Others	● Assisted in successful execution of the speaker sessions at the E-Summit conducted by E-Cell (2020) ● Completed a biathlon that consisted of 5 kms running and 7 kms cycling (2019) ● Successfully completed a year-long guitar training under National Sports Organization (2018-19) ● Participated in Dainik Bhaskar's National Level Newspaper Making Competition (2014)